

What is the type signature of learning?

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Learning builds agents

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- Today's agents are built using learning.
- Future agents probably will be too.
- We need learning processes that produce safe agents.

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- Objection 2: What does the update guarantee?

Objection 1: computation

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Can we compute it exactly with $\text{poly}(n)$ queries?

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Can we compute it exactly with $\text{poly}(n)$ queries?

No. Calculating the denominator requires 2^n queries.

Objection 2: guarantees

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$$\min_a \mathbb{E}_{\theta \sim p} L(a, \theta)$$

Minimize prior-averaged loss.

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Worst case

$$\min_a \max_{\theta} L(a, \theta)$$

Minimize the largest loss.

A small average can hide a large failure

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Average loss

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Worst-case loss

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Only then can average-case behavior approach the optimum for the true world.

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- A tractable learner has finite time and may need approximations.
- We cannot rely on infinite posterior concentration.
- Small Bayesian risk does not imply small worst-case loss.